

Inaccessibility to 911 and 311 Call Log Data and the Limitations it Creates for AI-Driven Hazard Detection: Case Studies of NYC Floods (2025)

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Urban flood hazards have been increasingly analyzed by AI softwares trained to pull information from a variety of sources, most of which are news articles, NOAA forecasts, and other external sources. However, despite the good documentation of floodings and the widespread impacts that these sources cover, there is little to no incorporation of individual flood related calls that are logged in 311 and 911 datasets in AI outputs. Unfortunately, most AI-softwares do not have access to these databases in the first place. Oftentimes, this data is incredibly inaccessible without human aid in decoding, filtering, and feeding the relevant information to them. As a result, when asked to comment on trends in NYC flooding for a specific year, their responses are almost entirely based on major news articles. This creates limitations in AI-software's ability to accurately determine instances of flooding in NYC for a specific timeframe, as it will only be able to confidently speak on major floods while ignoring instances that did not make headlines. This leads to inaccuracies in analyzing flood trends across NYC, stating them to occur much less frequently than they actually do. It is especially important for these trends to be adequately recorded as climate change and global warming continue to impact current flood rates. We aimed to explore these limitations by asking three common AI-softwares (ChatGPT, Perplexity, and Microsoft CoPilot) of their existing knowledge of NYC floods and their patterns for the year of 2025, and reviewed where they pulled their information. We also explored where they pulled their information on two recent major flood events, October 30th and July 14th 2025. Then, we exported recent datasets from NYC OpenData, and analyzed ~7 million 911 incident records, along with ~6 million 311 incident records all from 2025. We used RStudio and ChatGPT to filter the datasets and make them readable. We narrowed the datasets to show us the calls made on July 14th and October 30th, in addition to any other calls marked as flooding. Data was filtered by date, borough, and keywords (e.g., "Water," "Sewer," "Rescue," "Flood"), to locate exactly how many calls were made in relation to these floods. We found that these calls are not present within 911 databases, and most of them are logged under 311 databases. After feeding the filtered data to AI-Softwares, they were more confident in noting trends, patterns, and instances of flooding in 2025 when they had real-time data to cross reference with their original sources. This emphasizes the importance of integrating multi-source data and improving the accessibility of these datasets to AI-Software responsible for making decisions about flooding in NYC. It would prove useful to avoid inaccurate AI-driven decisions and ensure the most accurate information is being used in pattern detection.